

Trending Research Topics for M.Tech Dissertation

AI + DevOps / Cloud / Kubernetes / Docker / Terraform / CI-CD / Monitoring — 2025–2026

Domain Definition: "AI-Driven DevOps" (also called AIOps / Intelligent DevOps) — Using Artificial Intelligence, Generative AI, and Large Language Models (LLMs) to enhance, automate, and optimize DevOps practices across the entire software delivery and operations lifecycle. This covers all DevOps tools: Kubernetes, Docker, Terraform, Ansible, CI/CD pipelines, Prometheus, Grafana, ELK stack, and more.

1 LLM-Based Kubernetes Misconfiguration Detection & Auto-Remediation

Medium

🔥 Very Hot Kubernetes LLM Security DevSecOps

Using LLMs (GPT, LLaMA, etc.) to automatically detect security misconfigurations in Kubernetes YAML files and auto-fix them using RAG (Retrieval-Augmented Generation). Existing rule-based tools (KubeLinter, Checkov) have limited coverage. LLMs can understand context and fix configs while preserving functionality.

🔍 Research Gap:

Static analysis tools can detect misconfigurations but the industry lacks automated solutions capable of **FIXING** these misconfigurations. LLMSecConfig achieved 94% success rate — room to improve with better prompting, fine-tuning, or multi-tool approaches.

DevOps Tools: Kubernetes, Docker, Helm, KubeLinter, Checkov | **AI Tools:** LLM API (GPT/LLaMA), RAG, Prompt Engineering

RESEARCH SOURCES

- 📄 LLMSecConfig: An LLM-Based Approach for Fixing Software Container Misconfigurations (arXiv, Feb 2025)
- 📄 KubeGuard: LLM-Assisted Kubernetes Hardening via Configuration Files and Runtime Logs Analysis (arXiv, 2025)
- 📄 GenKubeSec: LLM-Based Kubernetes Misconfiguration Detection, Localization, Reasoning, and Remediation (arXiv, 2024)

2 GenAI-Powered Infrastructure as Code (IaC) Generation Across DevSecOps Lifecycle

Medium

🔥 Very Hot Terraform Ansible GenAI IaC

Using Generative AI to automatically generate Terraform, Ansible, Kubernetes manifests, and Helm charts from natural language descriptions. The AI acts as an "Infrastructure Copilot" — you describe what you want in plain English, and it generates production-ready, security-validated IaC scripts.

🔍 Research Gap:

No prior work has proposed an end-to-end, lifecycle-aware approach that synthesizes and refines IaC by leveraging artifacts from each DevSecOps stage. Current tools generate code but don't validate security or integrate across the

full lifecycle.

DevOps Tools: Terraform, Ansible, Kubernetes YAML, Helm, CloudFormation | **AI Tools:** LLM (GPT/Claude), Prompt Engineering, RAG

RESEARCH SOURCES

- 📖 [Generative AI as an Infrastructure Copilot: Automating IaC Across the DevSecOps Lifecycle \(Springer, Feb 2026\)](#)
- 📖 [Terraform Latest Trends 2026: AI Integration in IaC \(Clanker Cloud, Apr 2026\)](#)
- 📖 [Top AI-Powered Tools for Infrastructure Management 2026 \(StackGen, May 2026\)](#)

3 AI-Driven CI/CD Pipeline Generation & Optimization from Natural Language

Easy-Medium

🔥 Very Hot CI/CD GitHub Actions Jenkins LLM

Using LLMs to automatically generate CI/CD pipeline configurations (GitHub Actions, Jenkins, GitLab CI) from natural language descriptions of the project. Also includes AI-based pipeline optimization — predicting build failures, reducing build times, and auto-detecting flaky tests.

🔍 Research Gap:

No existing solution integrates natural language interpretation, repository-aware analysis, multi-platform pipeline generation, and automated validation into a cohesive framework. Current tools improve existing pipelines but can't CREATE them from scratch.

DevOps Tools: GitHub Actions, Jenkins, GitLab CI, ArgoCD | **AI Tools:** LLM API, Code Analysis, NLP

RESEARCH SOURCES

- 📖 [AutoPipelineAI: Context-Aware CI/CD Pipeline Generation from Natural Language \(arXiv, Jun 2026\)](#)
- 📖 [A Review of Generative AI and DevOps Pipelines: CI/CD, Agentic Automation, MLOps Integration \(Preprints, Jun 2025\)](#)
- 📖 [How AI is Transforming DevOps in 2026 \(Softjournal, May 2026\)](#)

4 LLM-Based Log Anomaly Detection for Cloud-Native Systems Medium

🔥 Very Hot Monitoring Observability LLM ELK Stack

Using LLMs (BERT, GPT, LLaMA) to parse, understand, and detect anomalies in application/system logs from Kubernetes, Docker, and cloud environments. This replaces traditional regex-based log parsing with AI that understands log semantics. Research papers like LogGPT, RAGLog, LLMeLog, and CodeAD are actively being published.

🔍 Research Gap:

Most existing work focuses on general system logs. Very few studies target Kubernetes-specific event logs combined with application logs for holistic anomaly detection. Also, LLM-based approaches are data-hungry — semi-supervised/few-shot approaches are underexplored.

DevOps Tools: Prometheus, Grafana, ELK Stack, Fluentd, OpenTelemetry | **AI Tools:** LLM (BERT/GPT), RAG, Few-shot Learning

RESEARCH SOURCES

- 📖 [Awesome LLM-AIOps: Curated List of LLM + AIOps Research \(GitHub, 2025–2026\)](#)
- 📖 [LLM4Log: A Systematic Review of LLM-based Log Analysis \(arXiv, 2025\)](#)
- 📖 [LLM meets ML: Data-efficient Anomaly Detection on Unstable Logs \(arXiv, 2024\)](#)
- 📖 [CodeAD: Synthesize Code of Rules for Log-based Anomaly Detection with LLMs \(arXiv, 2025\)](#)

5 AIOps: AI-Powered Root Cause Analysis & Automated Incident Response

Medium

🔥 Very Hot

AIOps

SRE

Monitoring

Observability

When an incident happens in cloud infrastructure, AI automatically correlates metrics, logs, and traces to find the root cause and suggest/execute remediation — replacing the traditional "war room" debugging approach. AIOps market grew from \$8.91B (2024) to \$11.16B (2025) and projected to reach \$32.56B by 2029.

🔍 Research Gap:

Only 4% of organizations have reached full AIOps maturity. 49% are still piloting. The gap between "AI recommendations" and "AI autonomous action" is wide. Research needed on safe autonomous remediation with human-in-the-loop guardrails.

DevOps Tools: Prometheus, Grafana, PagerDuty, Datadog, OpenTelemetry, K8s | **AI Tools:** LLM Agents, Time-series ML, Causal Inference

RESEARCH SOURCES

- 📖 [Observability and AIOps in Cloud-Scale DevOps \(ResearchGate, Nov 2025\)](#)
- 📖 [A Survey of AIOps in the Era of Large Language Models \(ResearchGate, Jun 2025\)](#)
- 📖 [AIOps: AI-Driven IT Operations and the Rise of Autonomous Infrastructure \(Zylos Research, Feb 2026\)](#)
- 📖 [AI Observability for LLM Systems: Multi-Layer Analysis — includes AIOpsLab Framework \(arXiv, 2025\)](#)

6 AI-Enhanced Self-Healing Kubernetes Clusters Medium

Kubernetes

Self-Healing

AI Operators

Automation

Extending Kubernetes' native self-healing (restart crashed pods) with AI that predicts failures BEFORE they happen, identifies root causes, and auto-remediates — creating truly autonomous clusters. Uses anomaly detection on metrics + reinforcement learning for remediation policies.

🔍 Research Gap:

Kubernetes' built-in self-healing is reactive and limited. AI integration evolves it into predictive, adaptive, and autonomous functionality, but end-to-end frameworks combining detection + prediction + remediation are still rare in literature.

DevOps Tools: Kubernetes, Prometheus, Custom Operators, Helm | **AI Tools:** Anomaly Detection, Time-series Forecasting, RL

RESEARCH SOURCES

- 📖 [AI-enhanced Self-Healing Kubernetes for Scalable Cloud Operations \(WJAETS, Nov 2025\)](#)
- 📖 [Towards Autonomous Kubernetes: A Framework for AI-Driven Operations \(IJISRT, 2026\)](#)
- 📖 [2026 Kubernetes Playbook: AI at Scale, Self-Healing Clusters \(Fairwinds, Dec 2025\)](#)

7

AI-Powered Container Image Vulnerability Detection & DevSecOps

Medium

Docker

Security

LLM

CI/CD

DevSecOps

Using AI/LLMs to scan Docker container images for vulnerabilities, analyze Dockerfiles for security best practices, and integrate into CI/CD pipelines for continuous security. AI reduces false positives (one case: 22,333 alerts reduced to 440 prioritized issues) and generates auto-fix suggestions.

🔍 Research Gap:

38% of IT security teams report challenges monitoring container workloads across distributed environments. LLM integration in CI/CD for security is emerging but lacks standardized evaluation benchmarks and real-world deployment studies.

DevOps Tools: Docker, Trivy, Clair, CI/CD Pipelines, SBOM tools | **AI Tools:** LLM embeddings, Binary classifiers, NLP

RESEARCH SOURCES

- 📖 [Enhancing DevSecOps Through LLM Integration: A Pipeline-Centric Approach \(Springer, 2026\)](#)
- 📖 [Large Language Models and GenAI in Cybersecurity: Survey of Dual-Use Risks \(arXiv, 2025\)](#)
- 📖 [AI in DevSecOps: Must Read for 2026 \(Practical DevSecOps, Jun 2026\)](#)
- 📖 [Container Security Vulnerabilities Explained: Image, Runtime, K8s, Supply Chain \(AI EdgeLabs, Jul 2026\)](#)

8

AI-Assisted Kubernetes Resource Optimization & FinOps

Easy-Medium

Kubernetes

FinOps

Cost Optimization

AI

Using AI to right-size Kubernetes pod resource requests/limits, recommend optimal node types, and predict cloud spending. CPU utilization across K8s clusters was just 8% in 2025, and CPU overprovisioning rose from 40% to 69%. AI can analyze historical usage patterns and recommend optimal configurations.

🔍 Research Gap:

Organizations are paying for infrastructure that their workloads don't even request, and the trend is accelerating. AI-driven FinOps that combines resource right-sizing with cost prediction is still largely commercial (CAST AI) — open-source research frameworks are lacking.

DevOps Tools: Kubernetes, Prometheus, Grafana, Cloud Cost APIs | **AI Tools:** Time-series Analysis, Recommendation Systems

RESEARCH SOURCES

- 📖 [2026 State of Kubernetes Optimization Report \(CAST AI, Jun 2026\)](#)
- 📖 [2026 Container Predictions — AIOps-driven FinOps \(DEVOPSDigest, 2026\)](#)
- 📖 [Safety-Gated Autoscaling: Multi-Layered Defense for K8s Resource Optimization \(arXiv, 2026\)](#)

9

Generative AI for DevOps Automation — A Comprehensive Survey

Easy

Survey Paper

GenAI

DevOps

All Tools

A systematic literature review (SLR) studying how Generative AI is being used across the entire DevOps lifecycle — code generation, infrastructure management, CI/CD, testing, monitoring, and incident response. This is a SURVEY-type dissertation (no prototype needed), covering 50+ research papers and analyzing trends, challenges, and future directions.

 **Research Gap:**

Existing reviews cover 2023–2025. A comprehensive, updated SLR covering 2024–2026 with a focus on Indian enterprise adoption, challenges in regulated industries, and comparison of commercial vs open-source AI-DevOps tools is missing.

DevOps Tools: All — K8s, Docker, Terraform, CI/CD, Monitoring, etc. | **AI Tools:** GenAI, LLMs, AIOps platforms

RESEARCH SOURCES

- A Review of Generative AI and DevOps Pipelines: CI/CD, Agentic Automation, MLOps (SSRN, Jun 2025)
- How AI is Transforming DevOps in 2026 (Softjournal, May 2026)
- AI-driven Kubernetes in Action: Exploring AI-Assisted K8s Operations (Cloud Native Now, May 2026)

10

AI Agents for Autonomous DevOps / SRE Operations (Agentic DevOps)

Medium-Hard

 Emerging

AI Agents

SRE

GitOps

Automation

Building AI agents that autonomously manage DevOps/SRE operations — an alert fires, the AI agent reads alert + context, proposes a Helm/Kustomize patch as a PR, a human approves, ArgoCD applies, and the alert resolves. This is the "human-in-the-loop GitOps" model where AI handles the toil while humans retain approval authority.

 **Research Gap:**

AIOps is evolving from "assistive" to "agentic" but safely deploying autonomous AI agents in production infrastructure is an open challenge. Trust, safety guardrails, and explainability of AI agent actions in critical infrastructure are underexplored.

DevOps Tools: ArgoCD, Helm, Kustomize, PagerDuty, Git, K8s | **AI Tools:** LLM Agents, LangChain, MCP, RAG

RESEARCH SOURCES

- AI Meets SRE in 2026: Autonomous Ops, Tools & Learning (yisusvii, Apr 2026)
- 2026 Observability Predictions: AIOps & AI Agents (APMdigest, Dec 2025)
- 2026 Observability & AI: Outlook and Trends for IT Leaders (LogicMonitor, Aug 2026)

Quick Comparison Table

#	Topic	DevOps Tools Covered	Difficulty	Prototype?
1	LLM-Based K8s Misconfiguration Detection	Kubernetes, Docker, Helm	Medium	Yes
2	GenAI-Powered IaC Generation	Terraform, Ansible, K8s YAML	Medium	Yes
3	AI-Driven CI/CD Pipeline Generation	GitHub Actions, Jenkins, GitLab CI	Easy-Medium	Optional

4	LLM-Based Log Anomaly Detection	ELK, Prometheus, Fluentd	Medium	Yes
5	AI Ops: AI Root Cause Analysis	Prometheus, Grafana, PagerDuty	Medium	Yes
6	AI Self-Healing K8s Clusters	Kubernetes, Operators, Helm	Medium	Yes
7	AI Container Security & DevSecOps	Docker, Trivy, CI/CD pipelines	Medium	Yes
8	AI K8s Resource Optimization / FinOps	Kubernetes, Prometheus, Cloud APIs	Easy-Medium	Yes
9	GenAI in DevOps — Survey (SLR)	All DevOps tools	Easy	No
10	AI Agents for Autonomous DevOps/SRE	ArgoCD, Helm, Git, PagerDuty	Medium-Hard	Yes